

NALLABATI VENKATA DURGA SAI SARAT CHANDRA

Target Role: Robotics & Controls Engineer Intern • Ather Energy

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EDUCATION & ACADEMIC PEDIGREE

Amrita Vishwa Vidyapeetham

August 2024 – Expected May 2028

B.Tech in Robotics & Artificial Intelligence (Cumulative GPA: 9.05 / 10.00, Top 5% Cohort)

Amritapuri, India

- **Academic Distinction:** AP POLYCET State Merit Rank: 497 (Top 0.5% statewide percentile across 100,000+ examinees).
- **Relevant Coursework:** Multivariable Calculus, Linear Algebra, Real-Time C++, ROS 2, Embedded Systems, Deep Learning, Classical Mechanics.

PEER-REVIEWED PUBLICATIONS & RESEARCH

- N. V. D. S. Sarat Chandra, M. Nunna, Asha C. A., P. C. Sreekumar, “Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity.” in *Proc. IEEE ICRM*, 2025. (First Author – Architected OpenCV 35+ FPS autonomy with 70% BOM cost reduction).
- N. V. D. S. Sarat Chandra, et al., “Physics-Guided Diffusion with Thermodynamic Awareness for 3D Molecular Generation.” in *IEEE IES*, 2026. (Finalist – Formulated SE(3) equivariant diffusion with Hamiltonian energy conservation; 3.2x faster convergence).
- N. V. D. S. Sarat Chandra, et al., “Domain-Aware Multi-Modal Industrial Diagnostics: Integrating Mechanical Heuristics with Conformal Uncertainty.” Submitted to *IEEE Transactions on Industrial Informatics*, 2026. (120k samples, Optuna ensemble, 95% coverage).

KEY ENGINEERING PROJECTS & SYSTEMS ARCHITECTURE

Pasupatha Autonomous Companion Robot Control Bus

Head of Embedded Systems, Club Mirai

ROS 2 Jazzy, STM32, ESP32, CAN 2.0B, C++ DLS Inverse Kinematics, HIL Testbed

- 50 Hz closed-loop control; <50us 5-DOF IK compute cycles (<0.4 deg repeatability).

Autonomous Mobile Robot (AMR) Edge Platform (IEEE ICRM 2025)

First Author & Lead Developer

ARM Cortex-A, OpenCV, Probabilistic Hough Transforms, Asynchronous UDP Sockets

- 35+ FPS lane boundary tracking; <15ms perception-to-actuation telemetry latency.

Evolution Edge: Self-Evolving Heterogeneous Neural Bridge

Team Lead, AMD Developer Hackathon

AMD Ryzen AI NPU, Vitis AI EP, ONNX Runtime, ROCm 6.x, INT8 PTQ, PEFT/LoRA

- Sub-28.4ms deterministic inference @ 8W TDP; offloaded tensor compute across integrated GPUs/NPUs.

HONORS, DISTINCTIONS & NATIONAL MERIT

- **ISRO-IIRS YUVIKA Space Science Scholar:** Selected among top nationwide students for residential aerospace and satellite telemetry training.
- **NAEST National Physics Test Winner:** Qualified Prelims and Finals in experimental and conceptual physics (IAPT / Prof. H.C. Verma).
- **MTSO National Olympiad Level 2 Qualifier:** National mathematics and algorithmic reasoning distinction.

TECHNICAL SKILLS & CORE COMPETENCIES

Programming Languages: C++ (C++17/C++20), Python (3.12), C (C99/C11), Bash, SQL, POSIX Sockets, Multi-Threading

Software Architecture & Design: Object-Oriented Design (OOP), Data Structures & Algorithms, REST APIs, Microservices, Asynchronous UDP/TCP Sockets

Robotics & AI Infrastructure: ROS 2 (Jazzy), CycloneDDS, OpenCV (35+ FPS), PyTorch, ONNX Runtime, Vitis AI EP, Zero-Copy Shared Memory

Tooling & CI/CD Systems: Linux (Ubuntu Server), Git/GitHub, Docker, CI/CD Pipelines, PyTest, GDB, Valgrind, CMake, Tectonic LaTeX